Digital Privacy, OPSEC & AI-Era OSINT

A hands-on lecture for CS/CE students



درس زبان تخصصی

استاد محمد صالح

2026

MOHAMMAD ABBASNEJAD

WHY SHOULD WE EVEN CARE?

Your Data Is Already a Product

~17B

location signals per day

Gravy Analytics claimed to curate signals from about 1B mobile devices. FTC allegations say the data could reveal sensitive visits.

01

Quick room check

Raise your hand if you used Incognito mode this week.

02

Reality check

Incognito hides local browsing history—not your activity from the site, ISP, or trackers.

03

Ask the room

What could a location trail reveal about a person’s routines?

A trail of ordinary pings can point to extraordinary parts of someone’s life.

WHY SHOULD WE EVEN CARE?

Five Ways Leaked Data Becomes Real Harm

Identity theft

Leaked identifiers can aid account recovery abuse.

Data brokers

Small clues combine into detailed profiles.

Social engineering

Personal details make scams more convincing.

Profiling

Patterns expose affiliations, habits, and vulnerabilities.

Irreversibility

Copies spread beyond your control; you can’t un-leak data.

OPSEC FOUNDATION

Privacy ≠ Anonymity — Know Your Threat Model

Privacy · who can access it?

- Control access to content and data
- Use end-to-end encryption
- Minimize what you disclose

Anonymity · who can link it to you?

- Separate personas and contexts
- Avoid cross-context logins
- Reduce identity-bearing clues

Signal with your real phone number can keep a message private—but does not make you anonymous.

NETWORK DEFENSE

VPN, Tor, Mullvad: Pick the Right Pipe



VPN

Shifts visible trust from your ISP to the provider. Choose audited, no-logs services—but understand you now trust that company.



Tor

Routes traffic through volunteer relays so no single point sees both origin and destination. Slower; not designed for everyday browsing.



Mullvad Browser

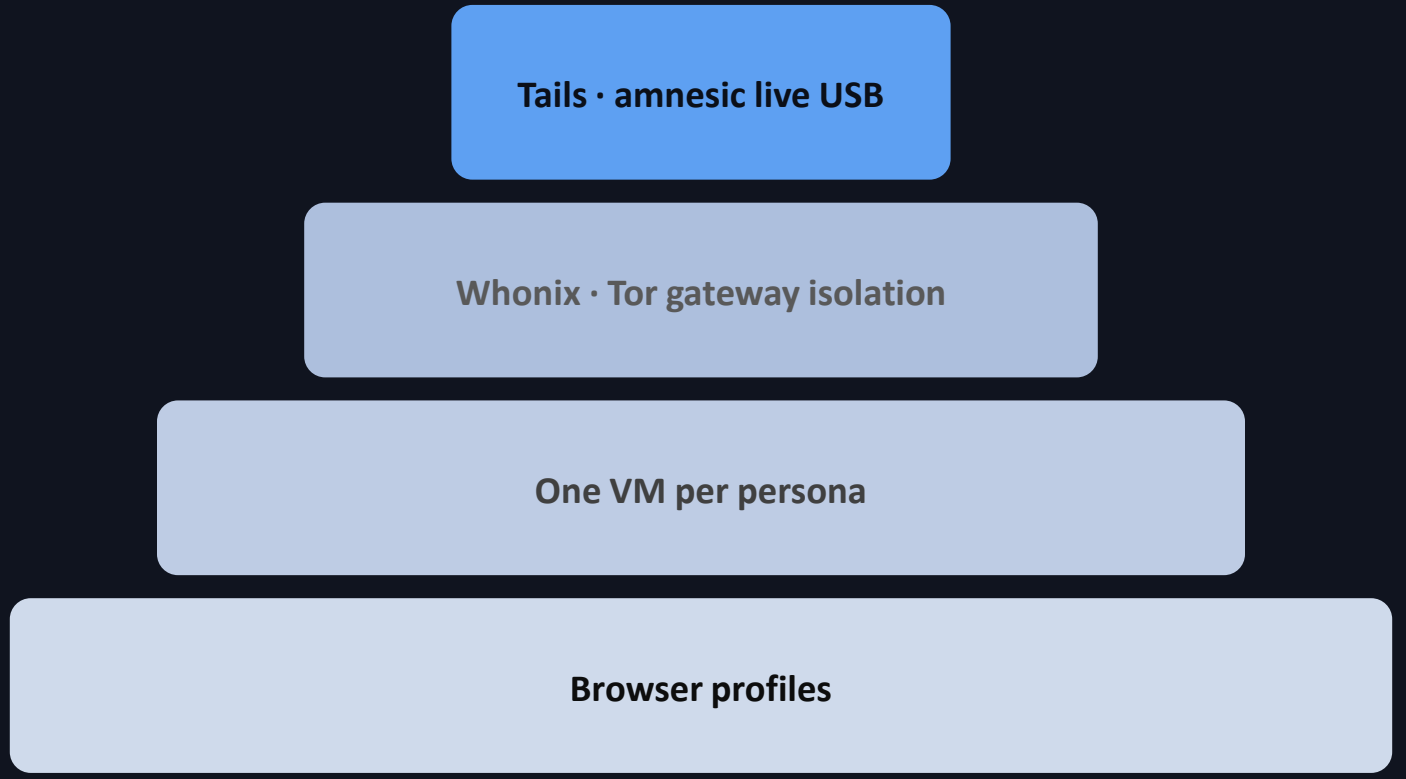
Built with the Tor Project to resist tracking and fingerprinting—without Tor's network latency. Pair it with a VPN or use it alone.



A VPN is not a magic shield — it changes who can spy on you.

ENVIRONMENT HARDENING

Compartmentalize: Never Mix Contexts



Leaves no trace on the host drive after shutdown.

Routes desktop traffic through Tor by design; its own docs stress tool limits.

Separate virtual machines keep work, personal, and research worlds apart.

Lightweight first step: keep sessions and cookies from cross-contaminating.

Think in terms of blast radius: if one identity burns, the rest survive.

FINGERPRINTING

Your Browser Fingerprint Gives You Away

What sites can read

- Canvas, WebGL, and audio rendering quirks
- Installed fonts and extensions
- Screen size, timezone, and user-agent

Why stacking defenses can backfire

- Tor Browser and Mullvad Browser aim for a uniform fingerprint
- Adding many unique extensions can make you stand out more
- Rule: blend in, don’t stand out
- Homework: test at **coveryourtracks.eff.org**

GENERATIVE AI

Treat Every AI Interaction as a Potential Data Disclosure

Prompts are data

What you paste can be logged, reviewed, or retained—check each provider’s policy, which varies by plan and workspace.

Telemetry persists

Even opt-outs and temporary chats can leave a short retention window for safety purposes.

Prompt injection

Untrusted content can steer an assistant to reveal or act on exposed data.

Local alternatives

Tools like Ollama run models on your own machine or an air-gapped system for sensitive work.



If you’re not paying for the AI, your data is the payment.

NEXT-GEN OSINT

AI Makes Cross-Platform Identity Correlation Faster

Linkability → reduce

Face search

PimEyes searches the open web for images containing a face; treat matches as leads, not proof.

Multimodal linking

A photo, voice clip, and public profile can reinforce one another.

Stylometry

Writing-style analysis can suggest a link, but it is not proof of identity.

Voice cloning

Verify unusual requests through a separate, trusted channel.

Exposure → reduce

CASE FILE 01 · PERSONAL OPSEC

When Ordinary Signals Expose Extraordinary Patterns

01

Strava · 2018

A public fitness heatmap made routes around military sites in Syria and Afghanistan visible.

02

Silk Road · identity bridge

A forum post and later job ad tied the same user to an email registered to Ross Ulbricht.

1 clue

can connect worlds

The lesson

The map exposed sensitive routines; in Silk Road, an old identity link helped investigators. Neither story required breaking cryptography.

CASE FILE 02 · BUSINESS DATA

A Prompt Can Become an Uncontrolled Data Channel

01

A normal work task

In 2023, Samsung semiconductor staff reportedly asked ChatGPT to debug code and summarize a meeting.

02

What crossed the boundary

Sensitive source code and meeting material entered a consumer AI service.

What changed

Samsung restricted staff AI use. The operational lesson: classify data before prompting and verify the service’s retention and training settings.

CASE FILE 03 · COUNTER-OSINT

Face Search Turns a Photo into a Search Key

One reused avatar

A small, public face image can become the bridge between an anonymous handle and real-world profiles.

Facial matches are clues—not proof.

Separate the signals

Use distinct avatars and usernames; strip EXIF metadata before posting. Do not treat a partial facial crop or sunglasses as reliable protection.

CASE FILE 04 · TRUST BOUNDARIES

VPNs and Extensions: The Privacy Paradox

The VPN myth

- A VPN shifts who can see your traffic; it does not grant immunity
- Trust moves from your ISP to the provider

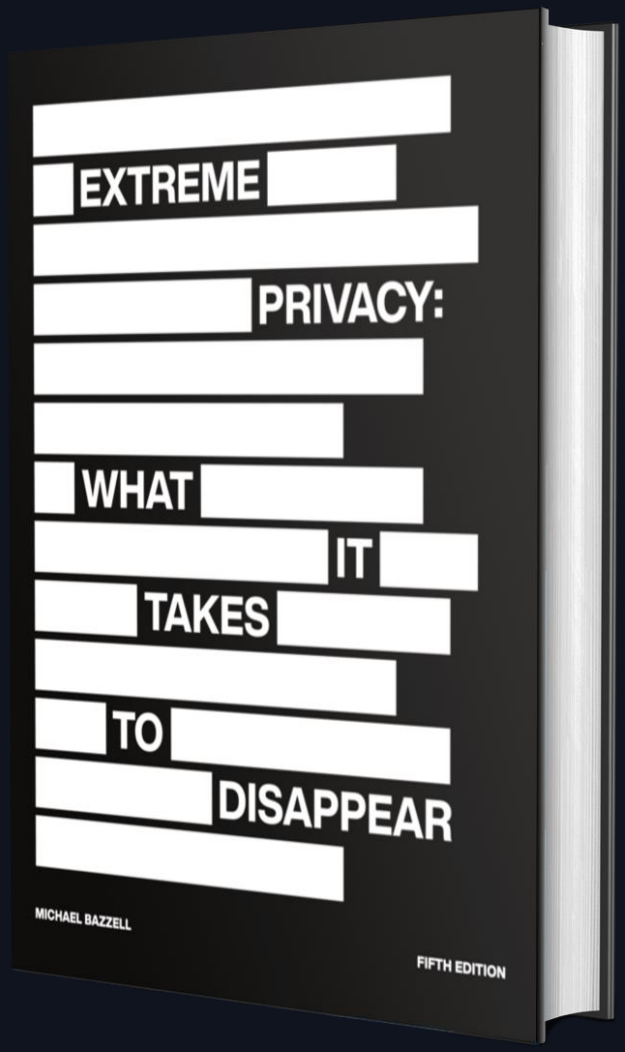
The extension paradox

- More extensions can make your fingerprint more unique
- Looking distinctive defeats blending in

Anonymity loves company — you cannot hide by looking unique.

My Book suggestion if you are interested

Michael Bazzell is a professional OSINT & Forensic investigator on behalf of the US Gov



Privacy and OPSEC Are Habits

Behavior beats tools

